Industry Task 1 - Employee Payroll Dashboard

Course: Programming in Java (CSA09) — Java AWT, Event Handling and Layout Managers

Task 1: Employee Performance & Payroll Dashboard

Problem Statement

A software company wants an Employee Performance and Payroll Management System built with Java AWT, letting HR enter employee details, process salary information, and generate a structured payroll report.

Components and Layout Manager Used

- **Layout:** `BorderLayout` for the frame (form panel in `NORTH`, report `TextArea` in `CENTER`, buttons in `SOUTH`), with a `GridLayout` inside the form panel to align the label/field pairs.
- **Components:** `Label`, `TextField` (ID, Name, Basic Salary, Working Days), `Choice` (Department, Designation), `CheckboxGroup` with `Checkbox` (Performance Rating: Excellent / Good / Average), `Button` (Generate Report, Reset), `TextArea` (report output).
- **Event Handling:** `ActionListener` attached to both buttons.

Business Rules Used

- Gross Salary = Basic Salary + (Basic Salary / Working Days assumed 30 x Working Days entered) — i.e. salary is pro-rated for days actually worked out of a 30-day month.
- Performance Bonus: Excellent = 20% of Basic, Good = 10% of Basic, Average = 0%.
- Net Salary = Gross Salary + Performance Bonus.

Java Source Code

```java
import java.awt.*;
import java.awt.event.*;

public class PayrollDashboard extends Frame implements ActionListener {

    TextField tfId, tfName, tfBasic, tfDays;
    Choice choiceDept, choiceDesig;
    CheckboxGroup ratingGroup;
    Checkbox cbExcellent, cbGood, cbAverage;
    TextArea reportArea;
    Button btnGenerate, btnReset;

    public PayrollDashboard() {
        setTitle("Employee Performance & Payroll Dashboard");
        setLayout(new BorderLayout(10, 10));

        // ---- Form Panel ----
        Panel formPanel = new Panel(new GridLayout(8, 2, 8, 8));

        formPanel.add(new Label("Employee ID:"));
        tfId = new TextField();
        formPanel.add(tfId);

        formPanel.add(new Label("Employee Name:"));
        tfName = new TextField();
        formPanel.add(tfName);

        formPanel.add(new Label("Department:"));
        choiceDept = new Choice();
        choiceDept.add("HR");
        choiceDept.add("Finance");
        choiceDept.add("Engineering");
        choiceDept.add("Sales");
        formPanel.add(choiceDept);

        formPanel.add(new Label("Designation:"));
        choiceDesig = new Choice();
        choiceDesig.add("Trainee");
        choiceDesig.add("Executive");
        choiceDesig.add("Senior Executive");
        choiceDesig.add("Manager");
        formPanel.add(choiceDesig);

        formPanel.add(new Label("Basic Salary:"));
        tfBasic = new TextField();
        formPanel.add(tfBasic);

        formPanel.add(new Label("Working Days:"));
        tfDays = new TextField();
        formPanel.add(tfDays);

        formPanel.add(new Label("Performance Rating:"));
        Panel ratingPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
        ratingGroup = new CheckboxGroup();
```

```java
        cbExcellent = new Checkbox("Excellent", ratingGroup, true);
        cbGood = new Checkbox("Good", ratingGroup, false);
        cbAverage = new Checkbox("Average", ratingGroup, false);
        ratingPanel.add(cbExcellent);
        ratingPanel.add(cbGood);
        ratingPanel.add(cbAverage);
        formPanel.add(ratingPanel);

        add(formPanel, BorderLayout.NORTH);

        // ---- Report Area ----
        reportArea = new TextArea(10, 50);
        reportArea.setEditable(false);
        add(reportArea, BorderLayout.CENTER);

        // ---- Buttons ----
        Panel btnPanel = new Panel(new FlowLayout());
        btnGenerate = new Button("Generate Report");
        btnReset = new Button("Reset");
        btnGenerate.addActionListener(this);
        btnReset.addActionListener(this);
        btnPanel.add(btnGenerate);
        btnPanel.add(btnReset);
        add(btnPanel, BorderLayout.SOUTH);

        setSize(500, 550);
        setVisible(true);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
                System.exit(0);
            }
        });
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == btnGenerate) {
            generateReport();
        } else if (e.getSource() == btnReset) {
            resetForm();
        }
    }

    private void generateReport() {
        String id = tfId.getText().trim();
        String name = tfName.getText().trim();
        String basicStr = tfBasic.getText().trim();
        String daysStr = tfDays.getText().trim();

        if (id.isEmpty() || name.isEmpty() || basicStr.isEmpty() || daysStr.isEmpty()) {
            reportArea.setText("Error: Please fill all required fields.");
            return;
        }

        double basic, workingDays;
        try {
            basic = Double.parseDouble(basicStr);
            workingDays = Double.parseDouble(daysStr);
        } catch (NumberFormatException ex) {
            reportArea.setText("Error: Basic Salary and Working Days must be numeric.");
            return;
        }

        if (basic <= 0 || workingDays <= 0 || workingDays > 31) {
            reportArea.setText("Error: Enter valid Basic Salary and Working Days (1-31).");
            return;
        }

        double grossSalary = (basic / 30.0) * workingDays;

        String rating = ratingGroup.getSelectedCheckbox().getLabel();
        double bonusPercent = 0;
        if (rating.equals("Excellent")) bonusPercent = 0.20;
        else if (rating.equals("Good")) bonusPercent = 0.10;

        double bonus = basic * bonusPercent;
        double netSalary = grossSalary + bonus;

        StringBuilder sb = new StringBuilder();
        sb.append("========= PAYROLL REPORT =========\n");
        sb.append("Employee ID     : ").append(id).append("\n");
        sb.append("Employee Name   : ").append(name).append("\n");
        sb.append("Department      : ").append(choiceDept.getSelectedItem()).append("\n");
        sb.append("Designation     : ").append(choiceDesig.getSelectedItem()).append("\n");
        sb.append("Working Days    : ").append((int) workingDays).append("\n");
        sb.append("Performance     : ").append(rating).append("\n");
        sb.append("-----------------------------------\n");
        sb.append(String.format("Basic Salary    : %.2f\n", basic));
        sb.append(String.format("Gross Salary    : %.2f\n", grossSalary));
        sb.append(String.format("Performance Bonus: %.2f\n", bonus));
        sb.append(String.format("NET SALARY      : %.2f\n", netSalary));
        sb.append("===================================\n");
```

```
            reportArea.setText(sb.toString());
        }

        private void resetForm() {
            tfId.setText("");
            tfName.setText("");
            tfBasic.setText("");
            tfDays.setText("");
            choiceDept.select(0);
            choiceDesig.select(0);
            ratingGroup.setSelectedCheckbox(cbExcellent);
            reportArea.setText("");
        }

        public static void main(String[] args) {
            new PayrollDashboard();
        }
    }
```

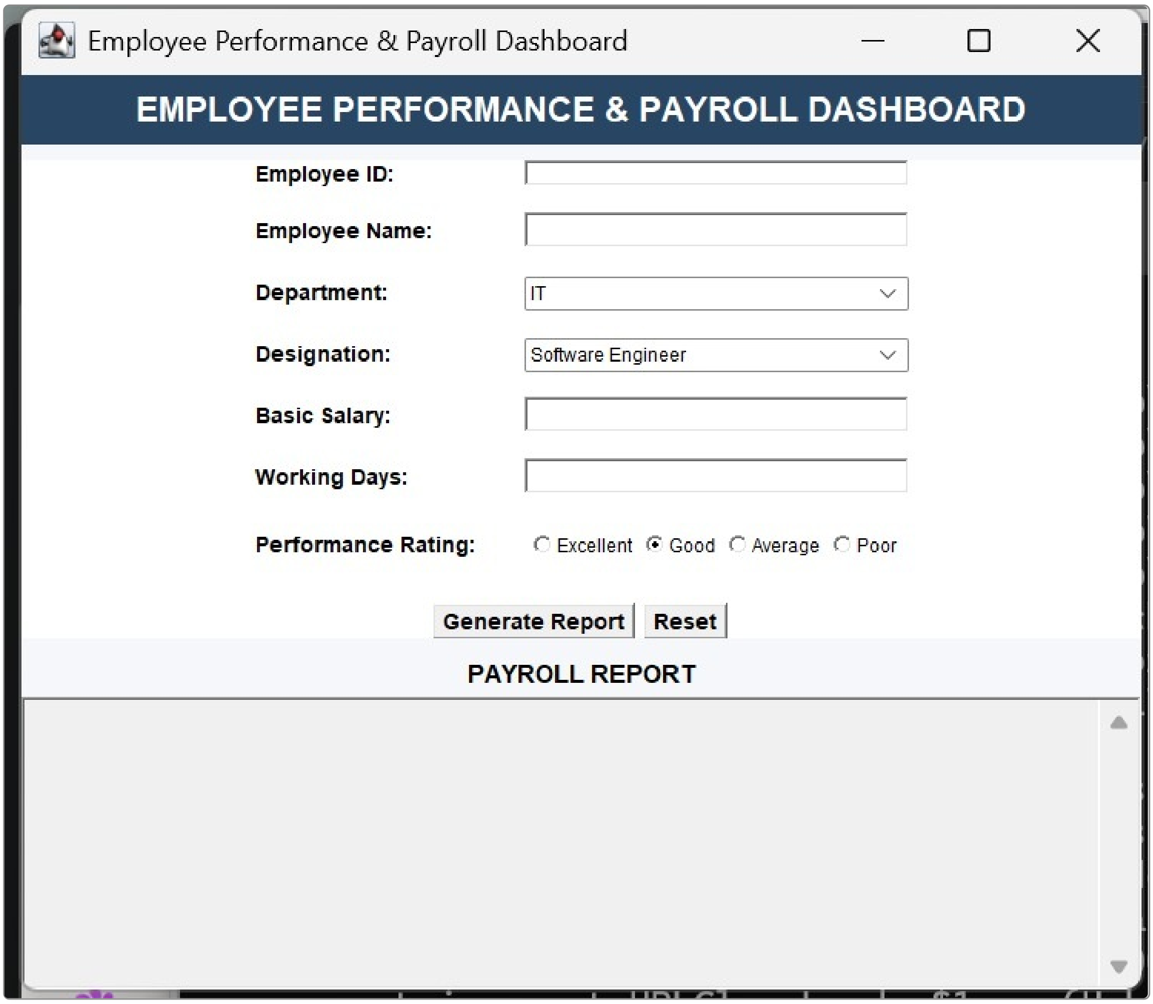
Sample Input / Output

Input: ID = E101, Name = Ananya Rao, Department = Engineering, Designation = Senior Executive, Basic Salary = 45000, Working Days = 28, Rating = Excellent

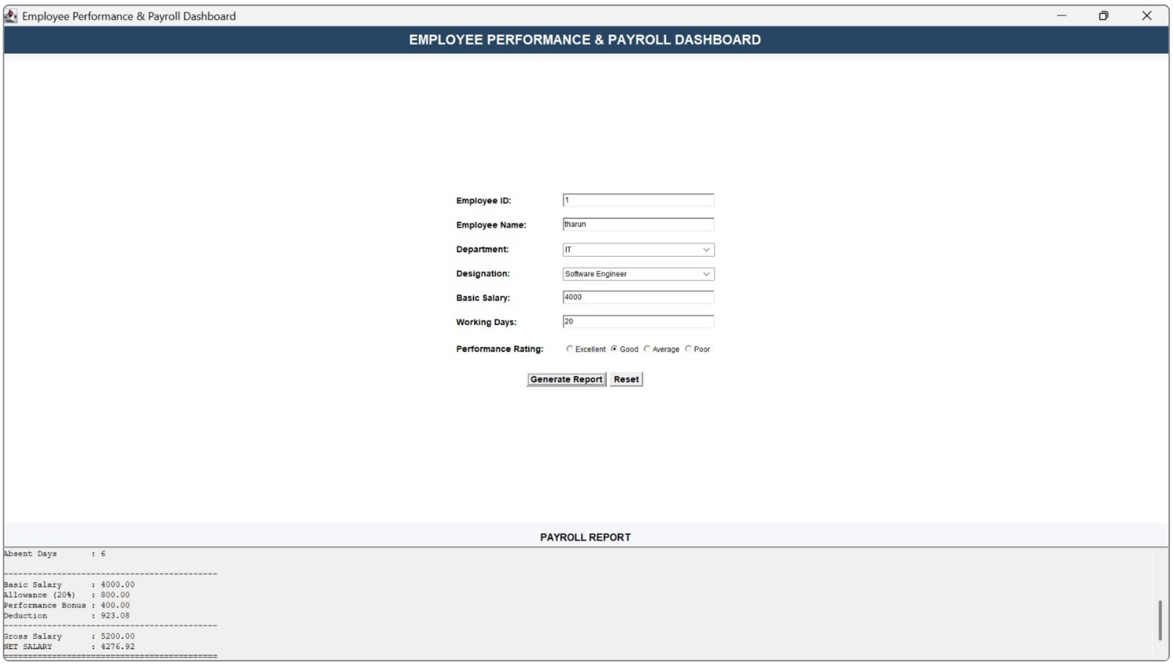
Output (in TextArea):

```
========= PAYROLL REPORT =========
Employee ID     : E101
Employee Name   : Ananya Rao
Department      : Engineering
Designation     : Senior Executive
Working Days    : 28
Performance     : Excellent
-----------------------------------
Basic Salary    : 45000.00
Gross Salary    : 42000.00
Performance Bonus: 9000.00
NET SALARY      : 51000.00
===================================
```

Screenshots of the Running Application



Employee Performance & Payroll Dashboard — blank input form



Employee Performance & Payroll Dashboard — after clicking "Generate Report"

Compilation & Execution

```
javac PayrollDashboard.java
java PayrollDashboard
```

Evaluation Rubric

Criterion	Weightage	Marks / Remarks
Problem Understanding	15%	
GUI Components and Layout	20%	
Event Handling	20%	
Functional Correctness	25%	
Input Validation and User Experience	10%	
Program Explanation / Viva	10%	

Student Instructions

1. The solution must be implemented in Java using the concepts covered in the laboratory syllabus.
2. Students should demonstrate compilation, execution, and output during the practical session.
3. Use meaningful variable names and appropriate comments.
4. Students must be able to explain the selected AWT components, layout manager, event handling mechanism, and processing logic.